

Computer Networks Lab Report: Network Traffic and Protocol Analysis

Course: Computer Networks Lab (Assignment 6)

Date: October 17, 2025

Topic: Network Traffic Capture and Protocol Analysis using Wireshark

1. Objective

The objective of this assignment was to capture and analyze live network traffic using Wireshark. This involved identifying various network protocols, interpreting packet-level details such as headers and fields, and understanding core communication patterns through filtering and analysis.

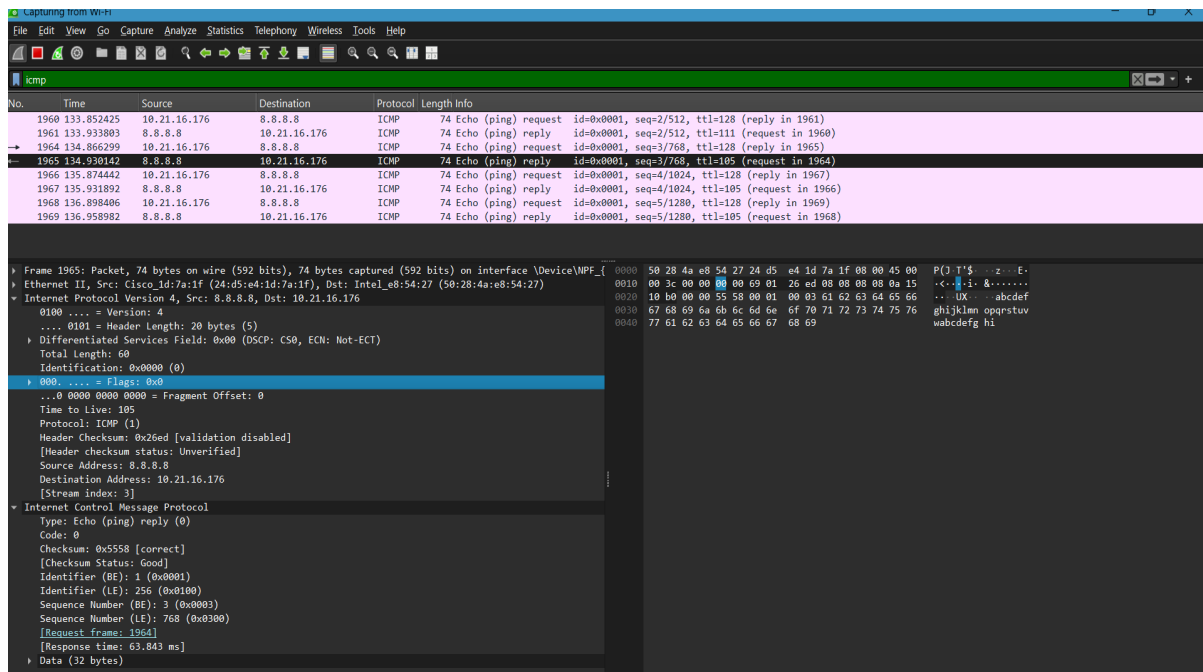
2. Task 1: Packet Capture

A 2-minute packet capture was initiated on the active Wi-Fi interface. During the capture, traffic was generated by pinging 8.8.8.8 and visiting several websites (<http://example.com>, <https://wikipedia.org>). The resulting data was saved as `capture_lab1.pcap`.

3. Task 2: Protocol Identification

Display filters were used to isolate specific protocols from the main capture file.

3.1. ICMP Traffic



The icmp filter successfully isolated the "Echo (ping) request" and "Echo (ping) reply" packets generated by the ping 8.8.8.8 command.

3.2. HTTP Traffic

The image shows a Wireshark packet capture of HTTP traffic. The top pane displays a list of packets, with packet 930 selected. The middle pane shows the details of packet 930, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
64	3.195461	10.21.16.176	49.44.50.19	HTTP	208	GET /connecttest.txt HTTP/1.1
189	8.944963	49.44.50.19	10.21.16.176	HTTP	241	HTTP/1.1 200 OK (text/plain)
627	21.787166	10.21.16.176	104.86.189.82	HTTP	208	GET /connecttest.txt HTTP/1.1
690	29.032444	104.86.189.82	10.21.16.176	HTTP	245	[TCP Retransmission] HTTP/1.1 200 OK (text/plain)
780	34.029253	10.21.16.176	49.44.50.10	HTTP	165	GET /connecttest.txt HTTP/1.1
791	34.970310	49.44.50.10	10.21.16.176	HTTP	241	HTTP/1.1 200 OK (text/plain)
930	48.185536	10.21.16.176	23.76.157.98	HTTP	208	GET /connecttest.txt HTTP/1.1
932	48.346813	23.76.157.98	10.21.16.176	HTTP	241	HTTP/1.1 200 OK (text/plain)
1283	75.148765	10.21.16.176	96.17.194.241	HTTP	208	GET /connecttest.txt HTTP/1.1
1287	75.261334	96.17.194.241	10.21.16.176	HTTP	241	HTTP/1.1 200 OK (text/plain)
2270	166.241533	10.21.16.176	203.153.36.180	HTTP	136	GET /ncc.txt HTTP/1.1
2272	166.295461	203.153.36.180	10.21.16.176	HTTP	205	HTTP/1.1 200 OK (text/html)
3004	192.636262	10.21.16.176	104.86.189.82	HTTP	208	GET /connecttest.txt HTTP/1.1
3010	193.068392	104.86.189.82	10.21.16.176	HTTP	241	HTTP/1.1 200 OK (text/plain)
3723	288.482156	10.21.16.176	49.44.50.10	HTTP	165	GET /connecttest.txt HTTP/1.1
3726	289.517318	49.44.50.10	10.21.16.176	HTTP	241	[TCP Retransmission] HTTP/1.1 200 OK (text/plain)
4919	495.914871	10.21.16.176	49.44.50.10	HTTP	208	GET /connecttest.txt HTTP/1.1
4948	495.995969	49.44.50.10	10.21.16.176	HTTP	241	HTTP/1.1 200 OK (text/plain)
4975	496.053544	10.21.16.176	23.55.108.9	HTTP	136	GET /ncc.txt HTTP/1.1
4988	496.027515	23.55.108.9	10.21.16.176	HTTP	205	HTTP/1.1 200 OK (text/html)

Frame 930: Packet, 208 bytes on wire (1664 bits), 208 bytes captured (1664 bits) on interface \Device\NPF...
Ethernet II, Src: Intel_e8:54:27 (50:28:a4:e8:54:27), Dst: Cisco_id:7a:1f (24:d5:e4:1d:7a:1f)
Internet Protocol Version 4, Src: 10.21.16.176, Dst: 23.76.157.98
... 0101 = Header Length: 20 bytes (5)
... Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 194
Identification: 0x75ef (30191)
... 010 = Flags: 0x2, Don't fragment
... 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.21.16.176
Destination Address: 23.76.157.98
[Stream index: 32]
Transmission Control Protocol, Src Port: 61194, Dst Port: 80, Seq: 1, Ack: 1, Len: 154
Hypertext Transfer Protocol

The http filter isolated the unencrypted web traffic from visiting `http://example.com`. The GET request packet is clearly visible.

3.3. DNS Traffic

No.	Time	Source	Destination	Protocol	Length	Info
169	8.877666	10.21.16.176	1.1.1.1	DNS	126	Standard query 0xfa71 A stream-production.avcdn.net OPT
175	8.944963	172.16.1.80	10.21.16.176	DNS	233	Standard query response 0xf499 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net CNAME a-0019.a.dns.azurefd.net CNAME
225	8.950467	8.8.8.8	10.21.16.176	DNS	205	Standard query response 0xf499 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net CNAME a-0019.a.dns.azurefd.net CNAME
271	8.967042	8.8.8.8	10.21.16.176	DNS	205	Standard query response 0xf499 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net CNAME a-0019.a.dns.azurefd.net CNAME
278	8.967042	1.1.1.1	10.21.16.176	DNS	375	Standard query response 0xfa71 A stream-production.avcdn.net CNAME stream-production.avcdn.net.akamaized.net CNAME a6143.dscd.akamai.
279	8.967042	172.16.1.80	10.21.16.176	DNS	233	Standard query response 0xf499 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net CNAME a-0019.a.dns.azurefd.net CNAME
293	9.416418	172.16.1.82	10.21.16.176	DNS	205	Standard query response 0xf499 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a-msedge.net CNAME a-0019.a.dns.azurefd.net CNAME
325	9.817922	1.1.1.1	10.21.16.176	DNS	375	Standard query response 0xfa71 A stream-production.avcdn.net CNAME stream-production.avcdn.net.akamaized.net CNAME a6143.dscd.akamai.
483	16.455192	10.21.16.176	8.8.8.8	DNS	91	Standard query 0xd2e5 SRV _ldap._tcp.dc._msdcs.iitp.ac.in
484	16.488441	10.21.16.176	8.8.8.8	DNS	83	Standard query 0xb72d A www.msftconnecttest.com
515	17.774146	10.21.16.176	172.16.1.80	DNS	81	Standard query 0x3ad0 A fb-unicast.msedge.net
524	18.056702	10.21.16.176	172.16.1.82	DNS	81	Standard query 0x3ad0 A fb-unicast.msedge.net
528	18.189823	10.21.16.176	172.16.1.80	DNS	71	Standard query 0x47e7 A chatgpt.com
532	18.462208	10.21.16.176	172.16.1.80	DNS	91	Standard query 0xd2e5 SRV _ldap._tcp.dc._msdcs.iitp.ac.in
533	18.462333	10.21.16.176	172.16.1.82	DNS	91	Standard query 0xd2e5 SRV _ldap._tcp.dc._msdcs.iitp.ac.in
534	18.462480	10.21.16.176	8.8.8.8	DNS	91	Standard query 0xd2e5 SRV _ldap._tcp.dc._msdcs.iitp.ac.in
535	18.479071	10.21.16.176	172.16.1.82	DNS	71	Standard query 0x47e7 A chatgpt.com
536	18.494141	10.21.16.176	172.16.1.80	DNS	83	Standard query 0xb72d A www.msftconnecttest.com
537	18.494155	10.21.16.176	8.8.8.8	DNS	83	Standard query 0xb72d A www.msftconnecttest.com
538	18.494154	10.21.16.176	172.16.1.82	DNS	83	Standard query 0xb72d A www.msftconnecttest.com
539	18.573619	172.16.1.80	10.21.16.176	DNS	103	Standard query response 0x47e7 A chatgpt.com A 104.18.32.47 A 172.64.155.209
549	18.631314	172.16.1.80	10.21.16.176	DNS	150	Standard query response 0xd2e5 No such name SRV _ldap._tcp.dc._msdcs.iitp.ac.in SOA ns5.iitp.ac.in
550	18.638545	172.16.1.82	10.21.16.176	DNS	140	Standard query response 0xd2e5 No such name SRV _ldap._tcp.dc._msdcs.iitp.ac.in SOA ns5.iitp.ac.in
551	18.638545	172.16.1.82	10.21.16.176	DNS	103	Standard query response 0x47e7 A chatgpt.com A 172.64.155.209 A 104.18.32.47

```

Protocol: UDP (17)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.21.16.176
Destination Address: 172.16.1.80
[Stream index: 2]
User Datagram Protocol, Src Port: 60666, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x47e7
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
[Response Type: 5301]
0000 24 d5 e4 1d 7a 1f 50 28 4a e8 54 27 08 00 45 00  $...z P( J T'..E
0010 00 39 0e c4 00 00 80 11 00 00 0a 15 10 b0 ac 10  .9.....
0020 01 50 ec fa 00 35 00 25 c8 5b 47 e7 01 00 00 01  .P...5 % [G.....
0030 00 00 00 00 00 00 07 63 68 61 74 67 70 74 03 63  .....c hatgpt c
0040 ef 6d 00 00 01 00 01                                om.....

```

The dns filter isolated the "Standard query" and "Standard query response" packets, which are responsible for resolving domain names (like example.com) into IP addresses.

4. Task 3: Detailed Packet Analysis

One packet from each of the three filtered protocols was selected for in-depth analysis. The key header details are recorded in the table below.

The screenshot below shows the detailed packet-layer view for the ICMP Echo Request, with

A	B	C	D	E	F
Protocol	Source IP	Destination IP	TTL	Packt Length	Flags
ICMP	8.8.8.8	10.21.16.176	105	74	0x0
DNS	10.21.16.176	172.16.1.80	128	71	0x0100
HTTP	10.21.16.176	23.76.157.98	128	208.00	0x2

the Frame, Ethernet, Internet Protocol, and ICMP layers expanded.

5. Task 4: Custom Filter Creation

A custom filter was created to isolate all packets originating from the host IP (10.120.44.171) and destined for standard web ports (80 for HTTP or 443 for HTTPS).

Filter String:

```
ip.src == 10.120.44.171 && (tcp.port == 80 || tcp.port == 443)
```

This filter successfully displayed all outgoing HTTP and HTTPS traffic, which consisted primarily of TCP and TLSv1.2 (Transport Layer Security) packets.

(Insert your screenshot Q4.png here)

6. Summary and Key Insights

This lab provided several key insights into practical network communication.

■ **Most Active Protocols:** An analysis of the capture's Statistics > Protocol Hierarchy menu revealed that TCP and UDP were the dominant transport protocols, carrying TLS (for HTTPS), DNS, and HTTP traffic.

■ **Suspicious Traffic:** No suspicious or anomalous traffic was observed during this controlled capture. All traffic was directly attributable to the actions performed (pinging and web browsing).

■ **Key Insights:**

1. DNS is a Prerequisite: The capture clearly shows that a DNS query (using UDP)

must be sent to resolve a domain name before an HTTP/HTTPS connection (using TCP) can be established.

2. HTTP vs. HTTPS Encryption: A major finding was the visible difference between HTTP and HTTPS. Traffic to `http://example.com` was unencrypted and readable (e.g., `GET /`). In contrast, traffic to `https://wikipedia.org` was encrypted as TLS, demonstrating how HTTPS secures web communication by making the data unreadable to an eavesdropper.

3. ICMP for Diagnostics: The ping command's use of ICMP Echo Request/Reply

A	B	C	D	E	F
Protocol	Source IP	Destination IP	TTL	Packt Length	Flags
ICMP	8.8.8.8	10.21.16.176	105	74	0x0
DNS	10.21.16.176	172.16.1.80	128	71	0x0100
HTTP	10.21.16.176	23.76.157.98	128	208.00	0x2

packets was clearly observed, providing a practical demonstration of this core network diagnostic tool.

7. Bonus Task: tshark Command-Line Analysis

The tshark command-line utility was successfully used to automatically extract all HTTP request URLs directly from the .pcap file.

Command Used:

```
tshark -r ...capture_lab1.pcap -Y http.request -T fields -e http.host -e http.request.uri
```

(Insert your screenshot bonus.png here)

8. Conclusion

This assignment provided valuable hands-on experience in capturing, filtering, and analyzing

live network traffic with Wireshark. The tasks successfully demonstrated the layered model of networking and the fundamental protocols (DNS, ICMP, HTTP, TCP/UDP) that power the internet.